Group 13

Analysis of Rail Demand Data for London

Module : MATH 6183

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4.37569597
5.33765642
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Content

01 Background

02 Data Clean & Feature Engineering

03 Data Visualization

04 PCA

05 Regression Modelling

06 Classification

07 Association

08 Insight of Strength & Limitation

Background

NUMBAT Dataset

From Transport for London (TfL)

This dataset is the most comprehensive statistical dataset on **railway usage** and **travel patterns** available in London.

Day

Mon/TueWedThu/Fri /Sat/Sun

Period

every 15-minute period throughout the traffic day

Line

Underground/Overground/Light Railway/TfL Rail / Elizabeth Line/Trams

Sample

Ticketing data from smartcards and gateline

Data Cleaning & Feature Engineering

General Cleaning

Input

- NBT22_Definitions_Published.xlsm
- NBT23MON_outputs.xlsx
- NBT23TWT_outputs.xlsx
- NBT23FRI_outputs.xlsx
- NBT23SAT_outputs.xlsx
- NBT23SUN_outputs.xlsx

01 — Clean Function

Aim :
Remove NA & duplicate values
Rename
---
lower case
replace blank with "_"

02 — Reshaped Data

Aim :
Converted timeband minutes into a long format

New Columns :
week_of_day,time

03 — Create New Columns

From :
NBT22_Definitions_Published.xlsm

New Columns:
timeband_description,
hour, timeband_hours

04 — Merged Same Sheet

Link_Loads&Link_Frequencies
Station_Entries&Station_Exits
Station_Boarders&Station_Alighters

05 — Combine All Files

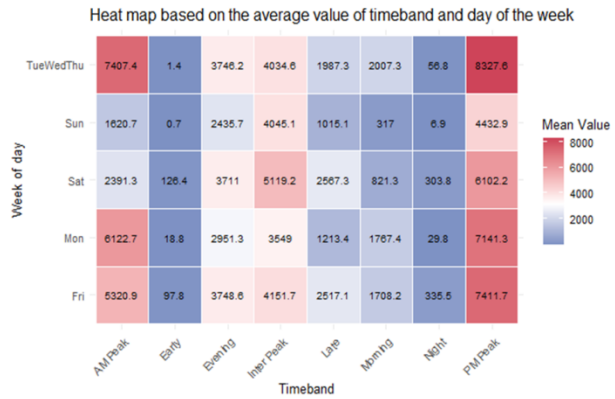
From:
MON -> SUN

Output

- Link_level.csv
- Station_level.csv
- Platform_level.csv
- Station_flows.csv
- Line_boarders.csv

Data Visualisation

Link_Level & Link_boarders

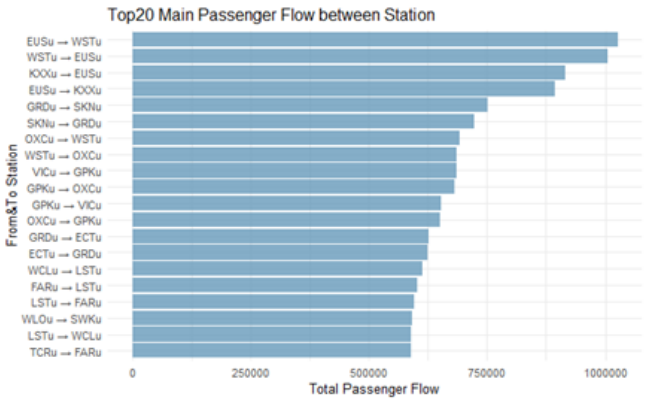


Same Traffic Pattern :

- The top five routes with the highest traffic were selected, and it was found that they generally increase during the AM and PM peak periods

Different Peak Heights :

- The Elizabeth Line and Northern Line show the highest peaks, suggesting they serve the busiest commuter corridors.



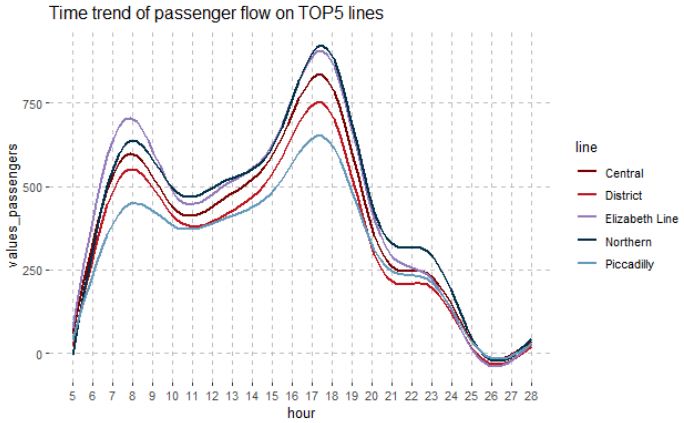
01 02 03

AM/PM Peak Traffic Patterns :

- Peak hour traffic on weekdays is significantly higher than on weekends
- Sunday has lower passenger flow than Saturday

Inter Peak Pattern:

- Saturday shows prominent inter peak traffic



Strong Commuter Corridors :

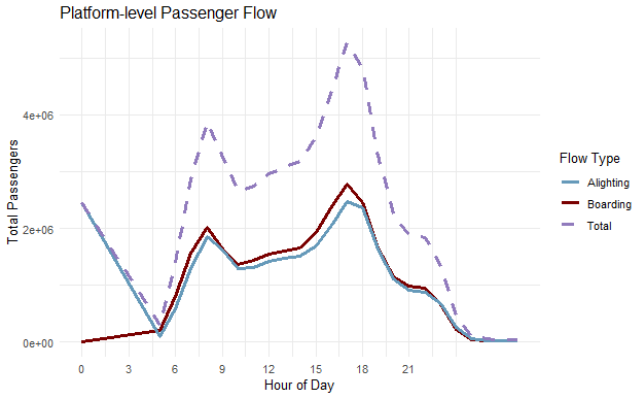
- Most high-flow links appear as bidirectional pairs, compliant with commuting mode.

Super Hub :

- Euston acts as a major super-hub, dominating the top flows in multiple directions.

Data Visualisation

Platform_Level & Station_Level & Station_Level

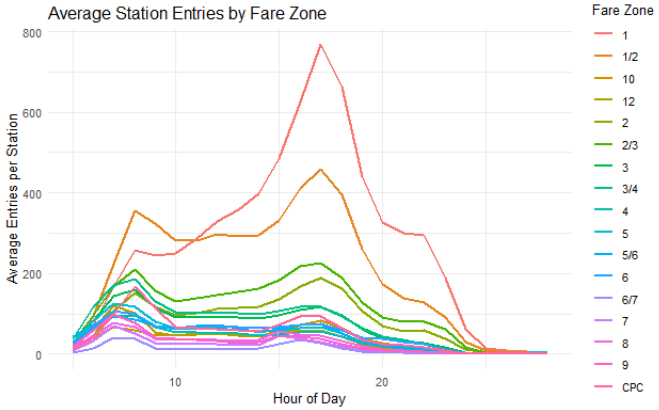


Weekday Pattern :

- Midweek (Tue–Thu) shows the strongest AM/PM peak with a sharp, concentrated spike.

Weekend Pattern :

- Both Saturday and Sunday lack a morning spike, flows build gradually into peak in afternoon.



04

05

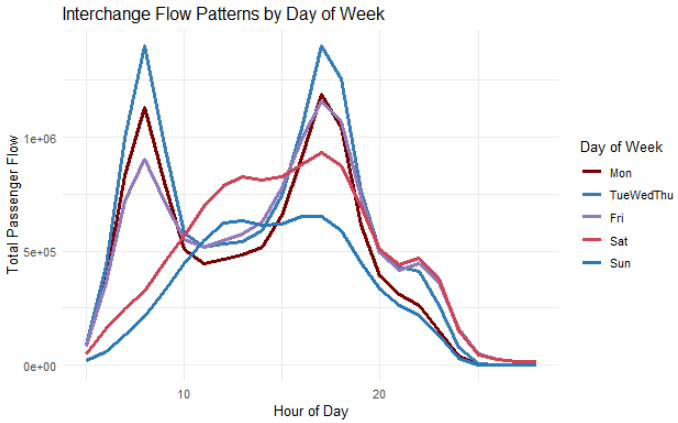
06

PM Peak :

- The PM peak is higher than the morning peak, driven by after-work and evening activity demand.

Boarding & Alighting overlap:

- Indicating balanced in–out flows typical of commuter travel



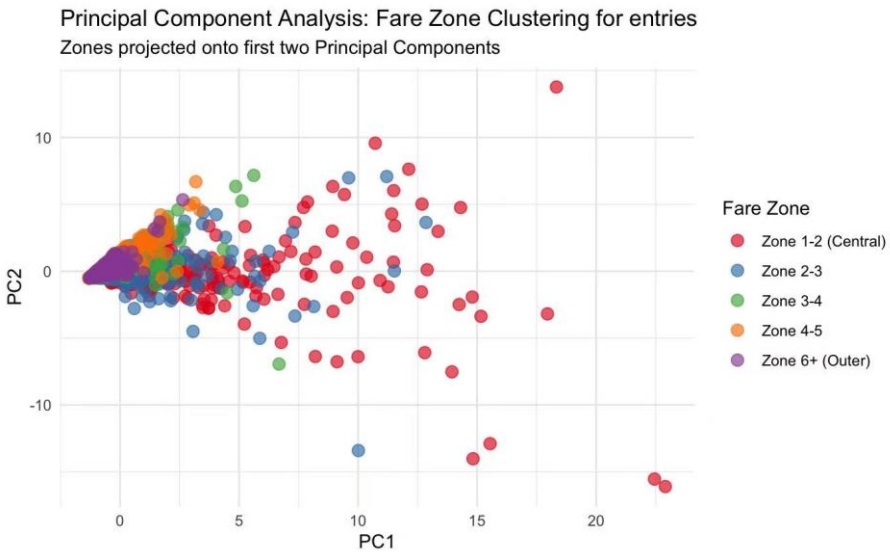
Fare Zone 1:

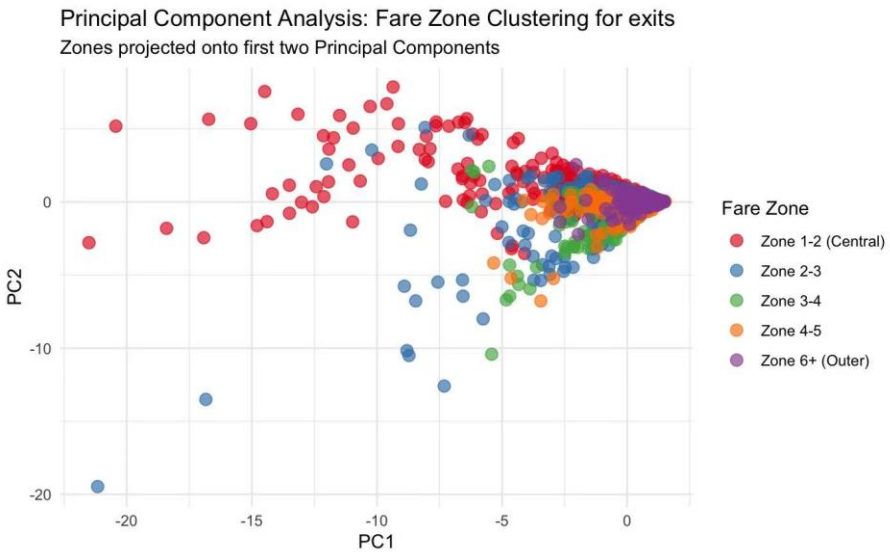
- Peak entries reach nearly 800, 2–4 higher than other zones
- Exhibit strong double peaks (AM + PM)

Outer zones :

- The peak at the top is weaker, and the distribution of travel time periods is more scattered.

PCA





Similarities:

- **Nearly Identical overall cluster structures :**
consistent entry and exit patterns across different zones over time.

- **Similar Outer Zone Patterns :**
Zones 4-5 and Zones 6+ almost overlap, suggesting that outer stations share highly similar temporal patterns.

Differences:

- **Greater Variance in Exits** :
PC1 is more stretched on the exit side, indicating higher variance in exit passenger flows.

- **Unique Behavior in Zones 1-2 :**
These zones show extreme variance and the strongest temporal behavior patterns compared to all other groups.

Predicting the number of passengers entering & exiting

Statistical Analysis Report

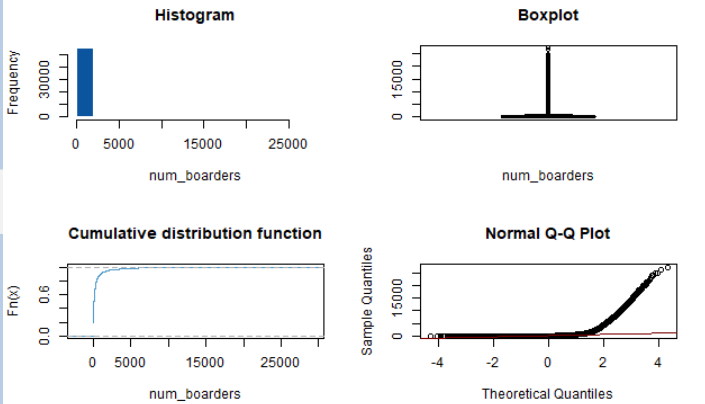
Histogram
Data Distribution

Box Plot
summary & outliers

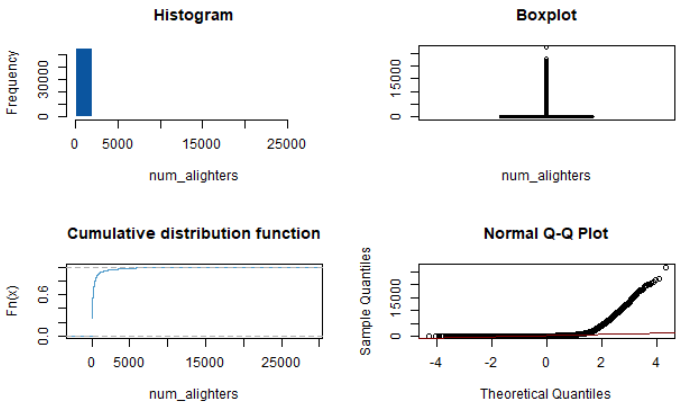
CDF
Cumulative distribution

Q-Q Plot
Data normality

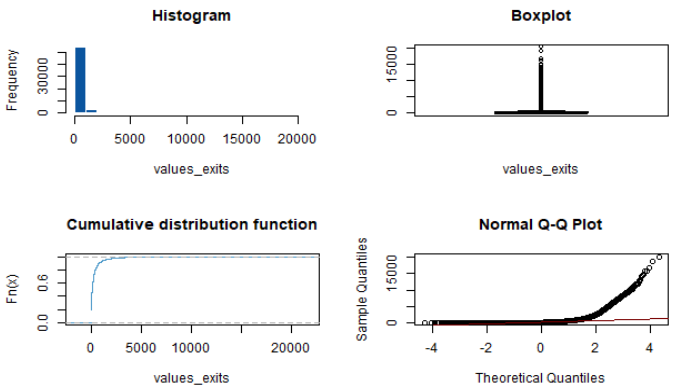
num_boarders



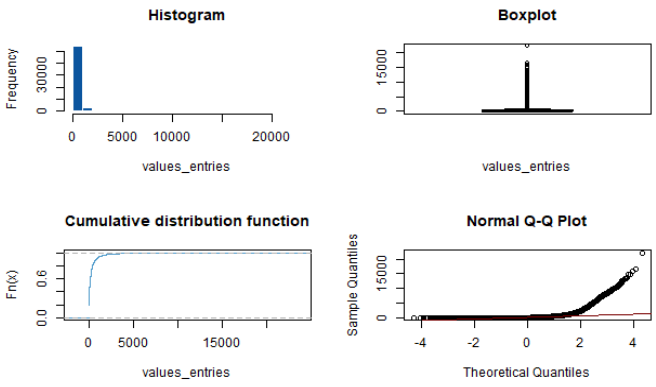
num_alighters



value_exists



value_entries



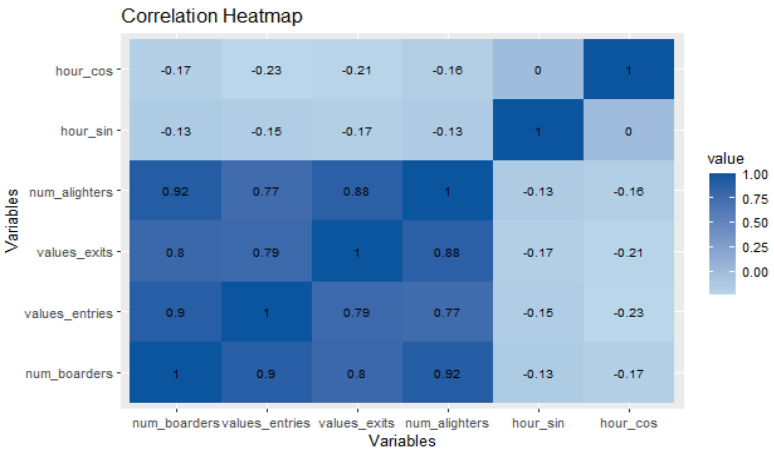
Predicting the number of passengers entering & exiting

Preprocessing steps

01 Add Variables

Variables Name	From Sheet Name
num_boarders	Station_Boarders
num_alighters	Station_Alighters
num_Lines	Link_Loads

02 Correlation Heat Map



03 Add Feature Variables

Old Variables	New Variables
hour	hour_sin
	hour_cos
	is_peak_hour
week_of_day	is_weekend
station_full	one-hot encoding
fare_zone	One-hot encoding

04 Remove Outliers

Before 58625 rows

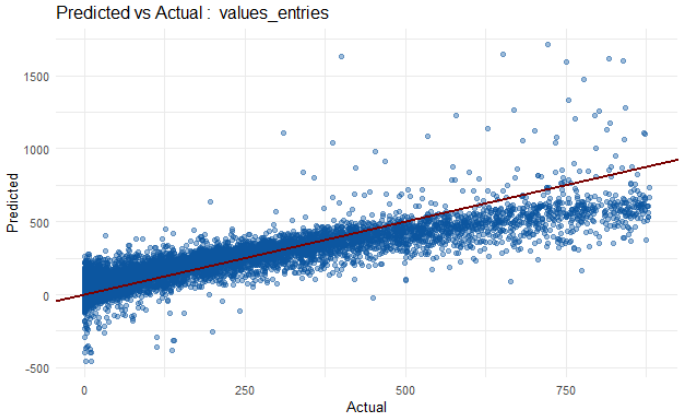
Method : IQR

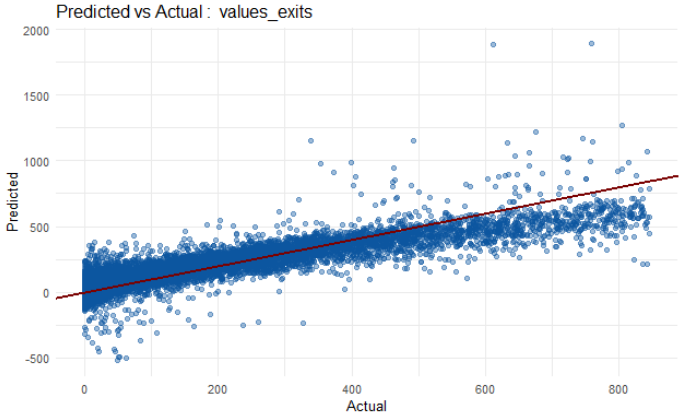
Removed : 28.2%

After 42088 rows

Predicting the number of passengers entering & exiting

Linear Regression





Key Metrics

R^2

Entries : 74.67%

Exits : 74.95%

MAE

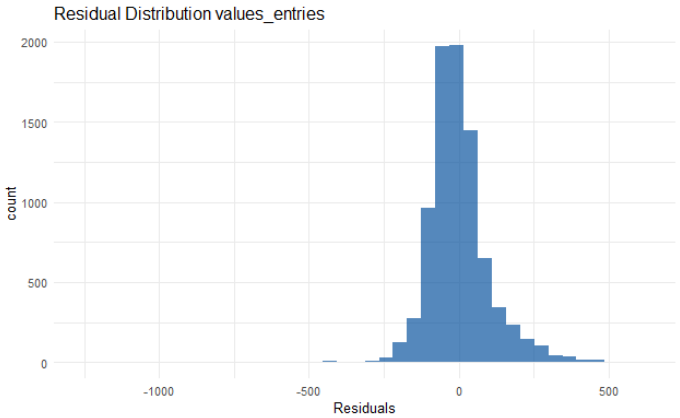
Entries : 73.18

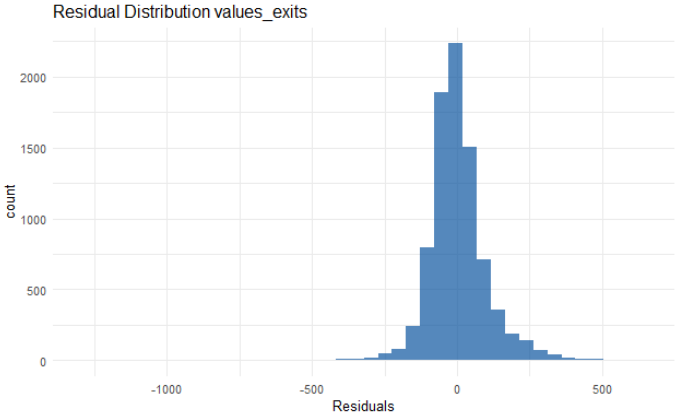
Exits : 69.28

RMSE

Entries : 104.80

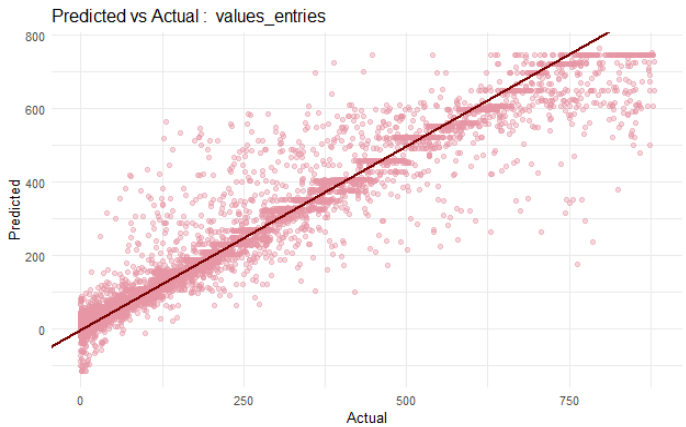
Exits : 99.81

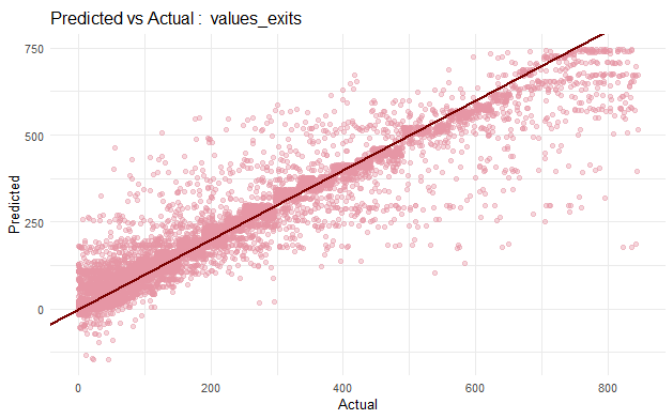




Predicting the number of passengers entering & exiting

XGBoost





Key Metrics

R^2

Entries : 93.19%

Exits : 88.79%

MAE

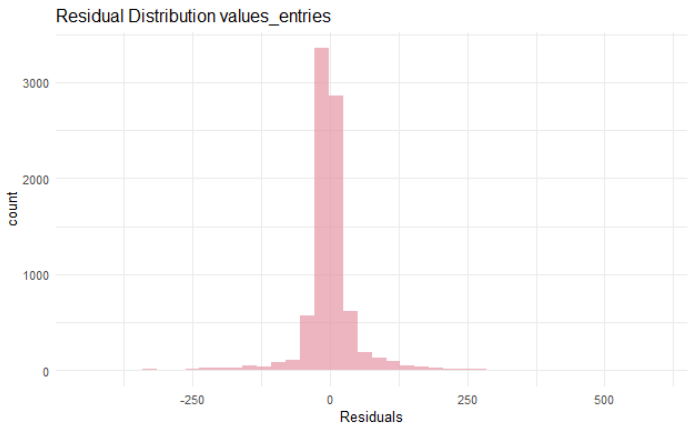
Entries : 28.38

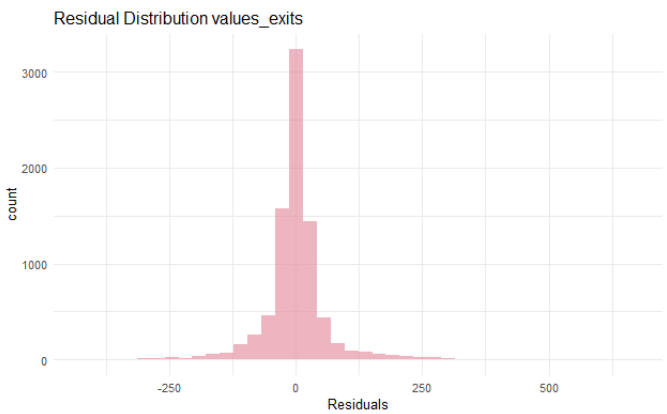
Exits : 38.02

RMSE

Entries : 55.44

Exits : 66.94





Predicting the number of passengers using links

Visualization of inks

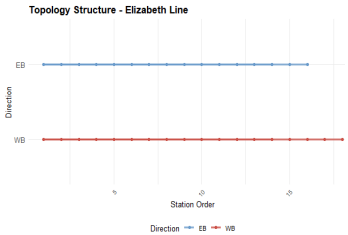
LINKS
1206

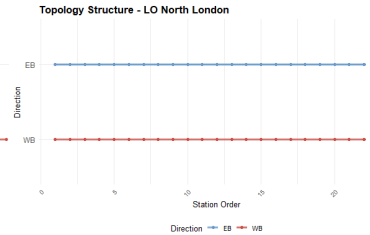
LINE
19

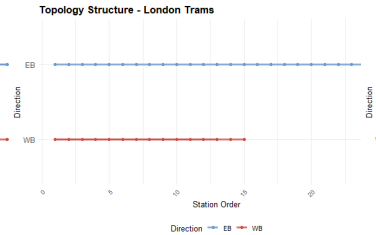
DIRECTION
8

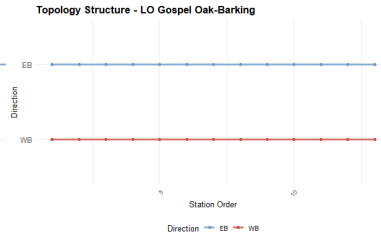
STATION
471

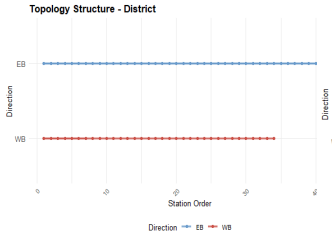
EB/WB

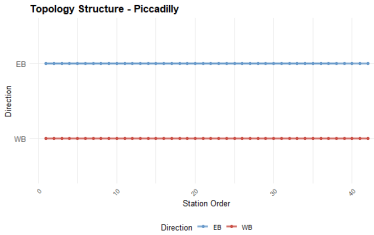


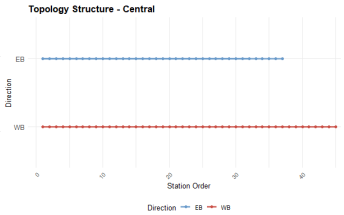


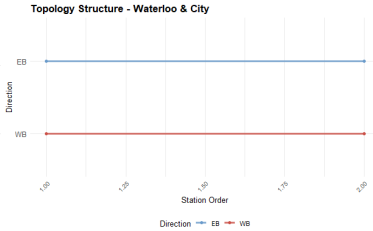




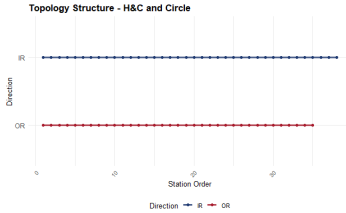




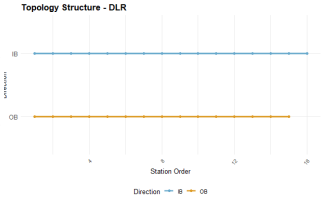




IR/OR

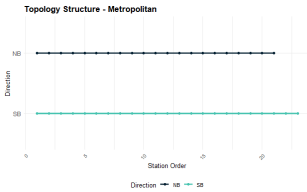


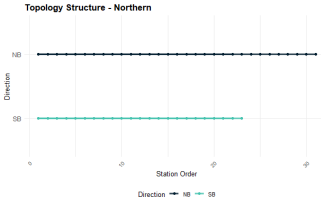
IB/OB

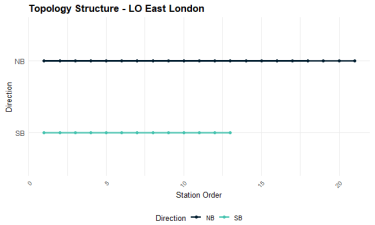


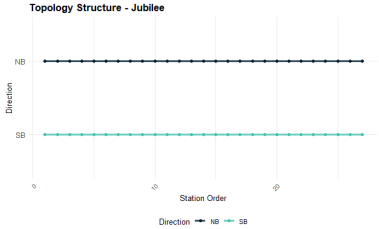
There are many lines in the NB/SB & EB/WB directions, only a portion of which are shown, while there are only two lines in the IB/OB IR/OR direction

NB/SB









Predicting the number of passengers using links

Preprocessing Steps

01 Add Position Variables

topology

avg_station_order	prev/next_station
min/max_station_order	num_sequences
main_start_station	station_order

Encoding

Line_scale_score	5	Big
	4	Bigger
	3	Middle
	2	Small
	1	smaller
Line_complexity_score	3	high
	2	meddle
	1	low

One-hot Encoding of direction

02 Add Time Variables

Encoding

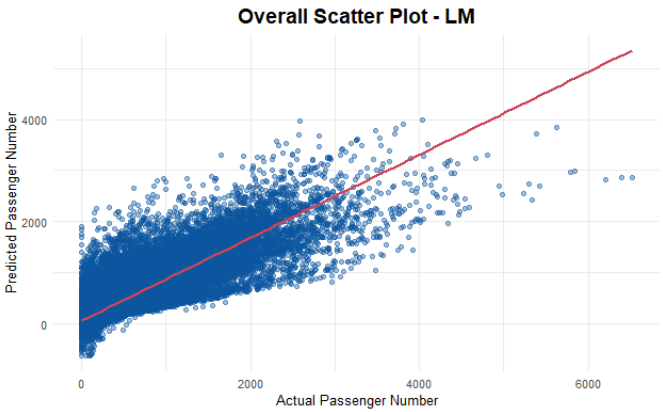
hour	hour_sin
	hour_cos
week_of_day	Is_weekend
timeband_description	one-hot encoding

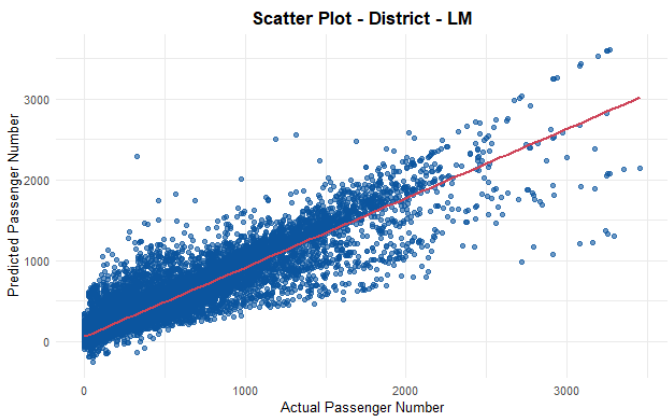
Function

P_lag	Provide the number of passengers before 15 mins/30 mins/1 hour/2 hour
P_roll	Provide the mean/max number of passengers during 1 hour/2 hour/3 hour Show the trend is increase or decrease

Predicting the number of passengers using links

Linear Regression





Key Metrics

Best Line : District

R^2

Total : 81.42%

Best : 86.91%

MAE

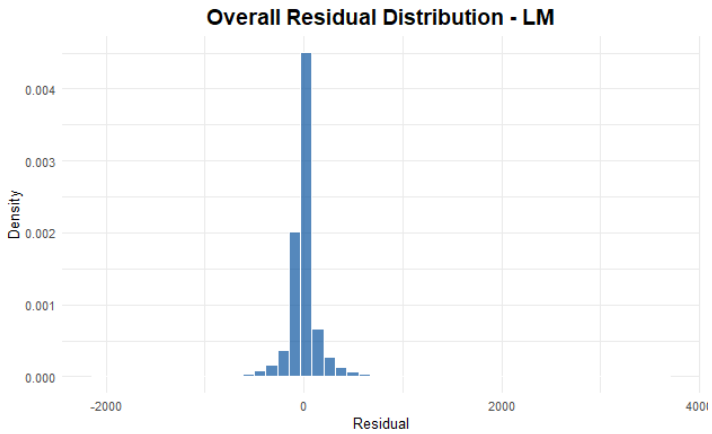
Total : 102.754

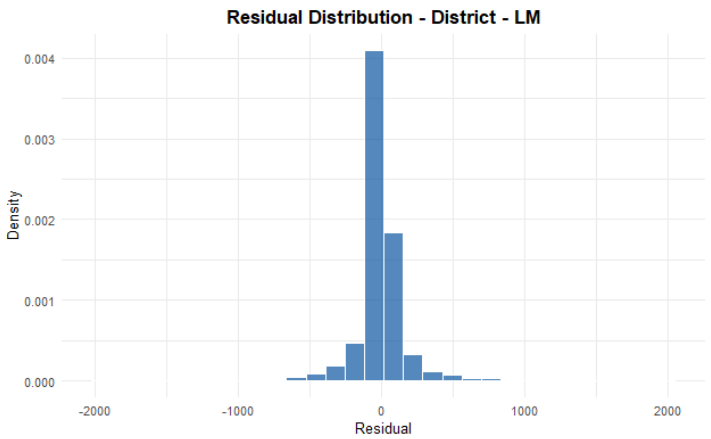
Best : 95.92

RMSE

Total : 200.31

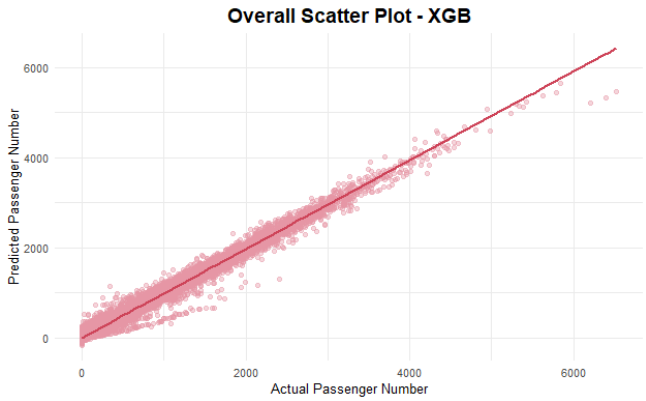
Best : 175.96

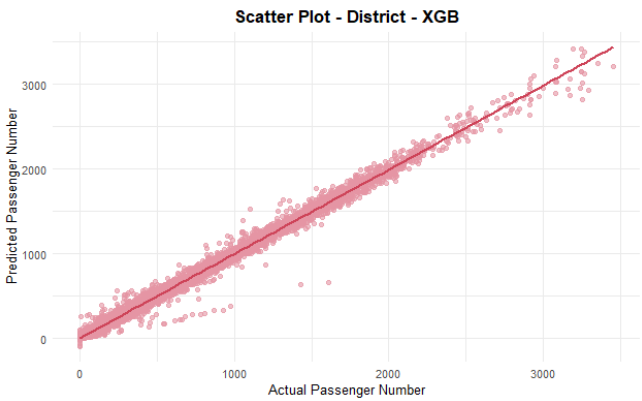




Predicting the number of passengers using links

XGBoost





Key Metrics

Best Line : District

R^2

Total : 99.28%

Best : 99.46%

MAE

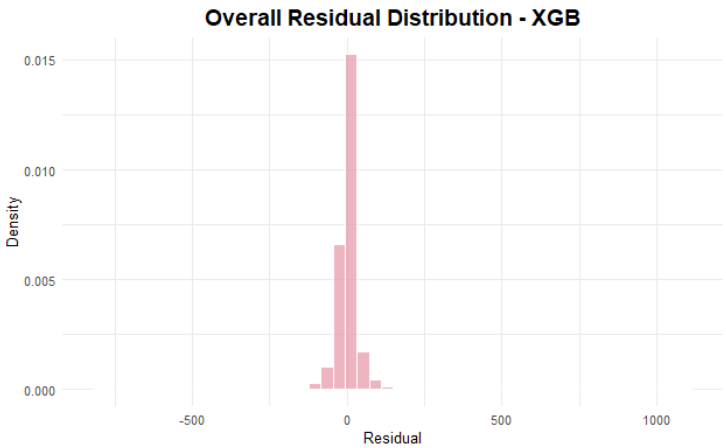
Total : 20.47

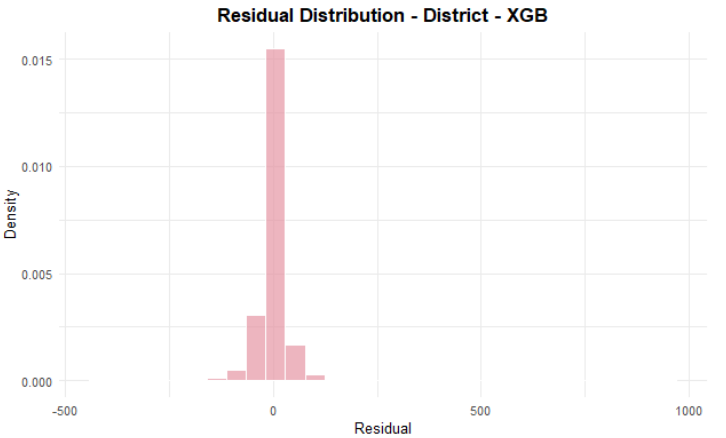
Best : 19.07

RMSE

Total : 39.52

Best : 35.75





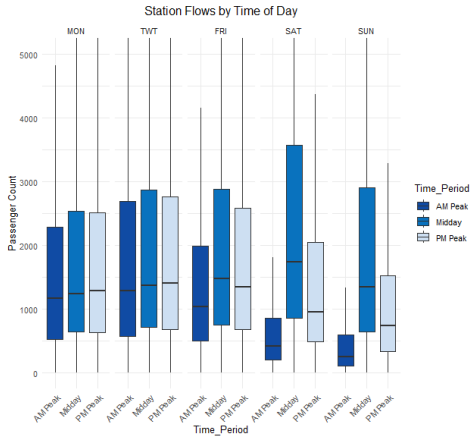
Thoughts on Classification Analysis

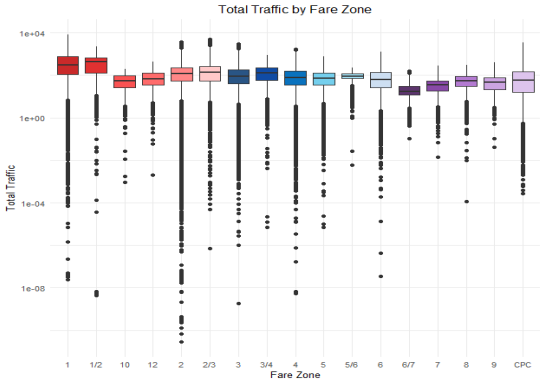
Critical Features

AM/PM Peak

A Clear Shift in Peak/Midday :
Weekdays show strong AM/PM peaks, while weekends shift to higher midday

01





Fare Zone

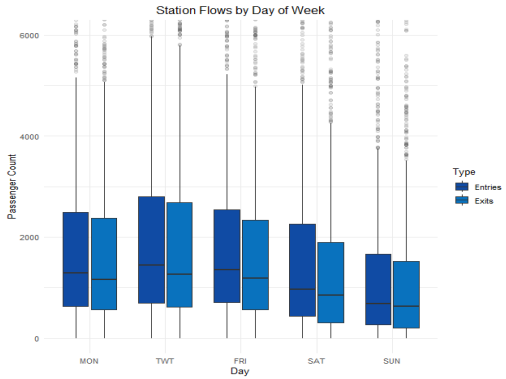
02

A strong hub effect of city center :
central zones (1) show the highest and most variable volumes

Weekend/Weekday

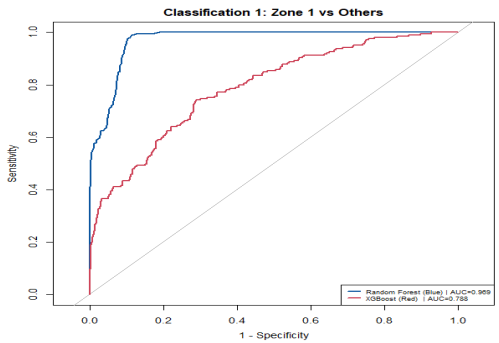
A Clear Commuter-driven Pattern :
Weekday passenger flows are significantly higher than weekend levels

03



Classification 01 : Total Traffic by Fare Zone

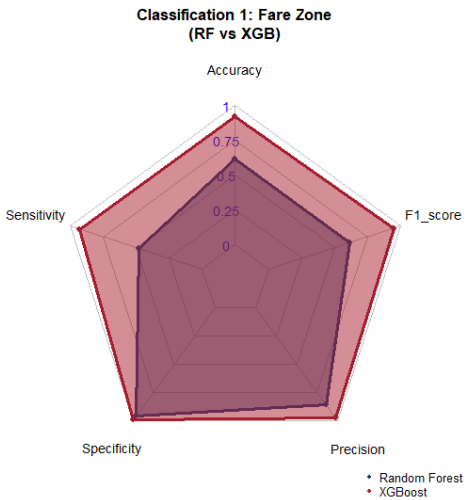
RF VS XGB



ROC Curve :
Classify the data into "Fare Zone 1" and "Not Zone 1" to distinguish the city center from the other areas. The ROC curve show how well it separates these two groups.

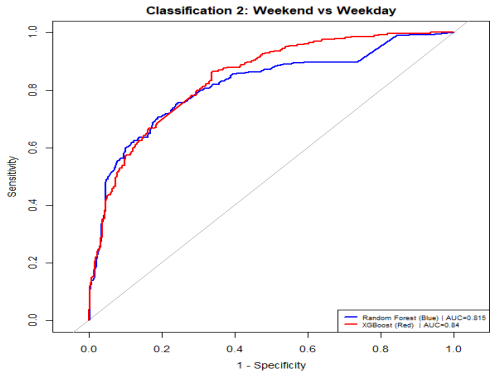
AUC :	RF : 0.969	XGBoost : 0.788

Accuracy :	RF : 0.624	XGBoost : 0.925
F1 Score:	RF : 0.617	XGBoost : 0.948
Sensitivity :	RF : 0.500	XGBoost : 0.932
Specificity :	RF : 0.947	XGBoost : 0.991
Precision :	RF : 0.821	XGBoost : 0.916



Classification 02 : Station Flows by Day of Week

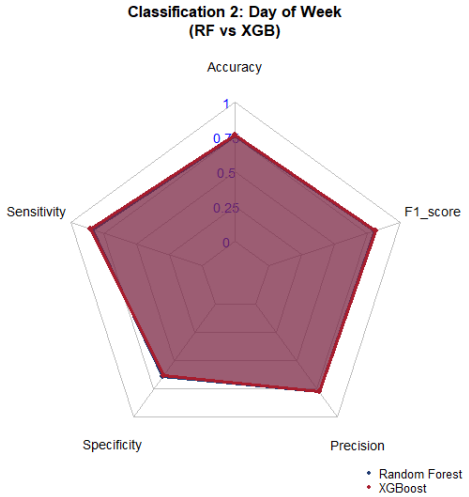
RF VS XGB



ROC Curve :
Classify the data into “Weekend” and “Weekday”.
The ROC curve show how well it separates these two groups.

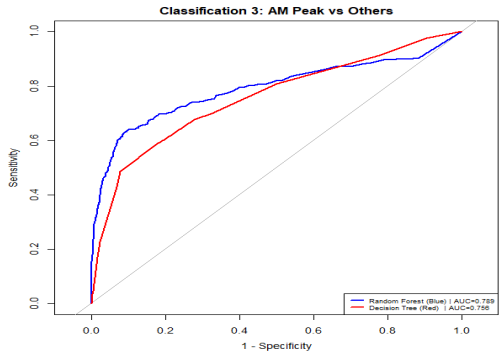
AUC :	RF : 0.815	XGBoost : 0.84

Accuracy :	RF : 0.767	XGBoost : 0.763
F1 Score:	RF : 0.813	XGBoost : 0.811
Sensitivity :	RF : 0.846	XGBoost : 0.851
Specificity :	RF : 0.648	XGBoost : 0.631
Precision :	RF : 0.783	XGBoost : 0.775



Classification 03 :Station Flows by Time of Day

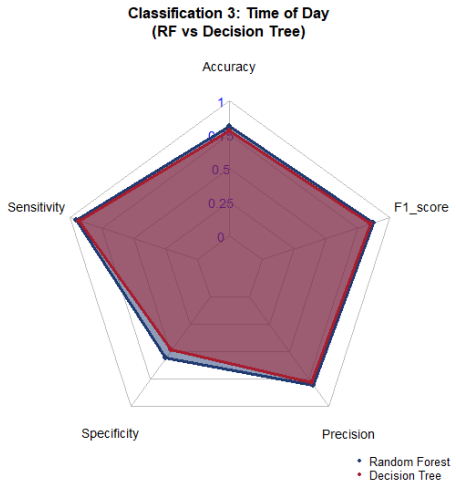
RF VS Decision Tree



ROC Curve :
Classify the data into "AM Peak" and "others" to distinguish the AM peak time from the other time during one day. The ROC curve show how well it separates these two groups.

AUC :	RF : 0.789	Decision Tree : 0.756

Accuracy :	RF : 0.814	Decision Tree : 0.776
F1 Score:	RF : 0.871	Decision Tree : 0.846
Sensitivity :	RF : 0.943	Decision Tree : 0.922
Specificity :	RF : 0.337	Decision Tree : 0.486
Precision :	RF : 0.809	Decision Tree : 0.781



Association

Methodology

Association for **Station as Transaction / Link as Transaction**

Define thresholds for **high/medium/low**

One-hot encoding - （from_station_full & to_station_full)

Analyzed Frequent item set with algorithmic controls

Filtered and analyzed insights based on **High lift** (strong association),**High support** (common pattern), **High confidence** (reliable), **High count** (stable across many rows)

Association : Station as a Transaction

Explore Links between Zone, Lines and Traffic by Timeband

Strong Interchange Effects

Interchange stations show strong associations with high passenger flows.

Inter Peak Timeband

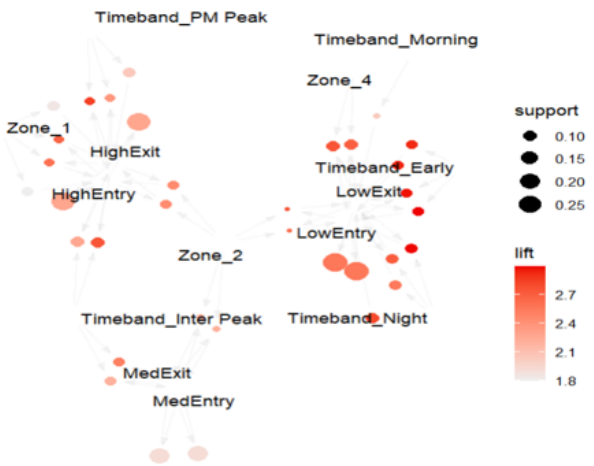
Inter-Peak accounts for **25%** of rules, indicating it is consistently the busiest period for station activity.

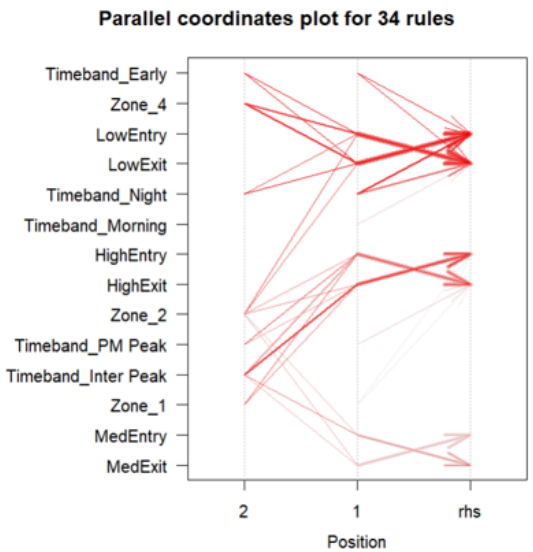
Clear Zone Effects

- **Zone 1:** High-traffic hubs with balanced flows (e.g. King's Cross) congestion management
- **Zone 2:** Strong commuter peaks
- **Zone 4:** Low and steady traffic

Directional Commuter Flows :

Rules capture **late-hour** outbound flows, showing a consistent directional pattern.





Top Zone by Lift	Top Zone by Confidence	Top Zone by Support	Most Associated Zone
Rule {LowExit, Zone_4} => {LowEntry}	Rule {HighEntry, Zone_1} => {HighExit}	Rule {LowExit, Zone_4} => {LowEntry}	Zone_2
Lift : 2.745	Confidence : 0.90	Support : 0.104	Frequency : 0.244

Association : Link as a Transaction

Examine Station Characteristics, Direction and Traffic Loads

H&C + Circle Line Exclusivity

Both inward (**IR**) and outward (**OR**) flows—across medium and high traffic levels—are always tied to the H&C and Circle line (confidence = **1.0**, lift ≈ **12.83**).

DLR Inward Bias

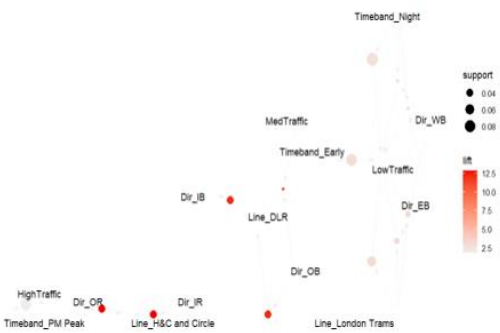
Both low and high traffic on DLR predict inward-bound (**IB**) direction (confidence ≈ **0.51**, lift ≈ **12.3**).

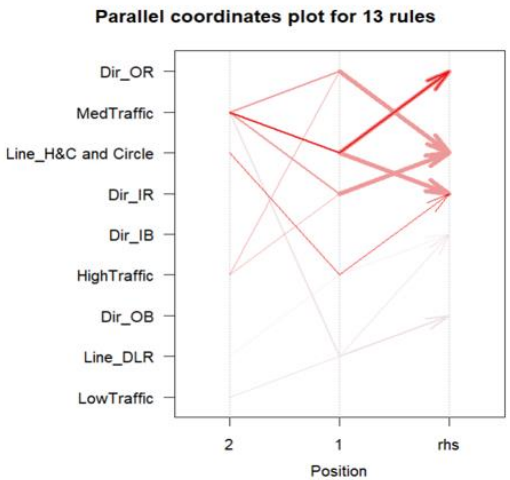
Top Zone by Lift	Top Zone by Confidence	Top Zone by Support	Most Associated Link
Rule {HighTraffic, Line_H&C and Circle} ⇒ {Dir_IR}	**Rule** {Dir_IR} ⇒ {Line_H&C and Circle}	**Rule** {Dir_IR} ⇒ {Line_H&C and Circle}	{Line_H&C and Circle} <=> {Dir_IR} & {Line_H&C and Circle} <=> {Dir_OR}
Lift 13.37	**Confidence** 1.00	**Support** 0.031	**Frequency** 22560

Traffic–Direction Interplay on H&C & Circle

High traffic on H&C and Circle predicts inward direction(**IR**)(confidence ≈ **0.52**, lift ≈ **13.37**), while medium traffic predicts outward direction(OR) (confidence ≈ **0.52**, lift ≈ **13.30**).

Traffic intensity influences directional bias. High loads skew inward, medium loads skew outward. This could inform directional scheduling and capacity allocation.





Insight of Modelling

Regression Classification and Association

Strengths

Regression

- **XGBoost**
  Captures complex nonlinear patterns, giving much higher predictive accuracy.
- **Linear Regression**
  Highly interpretable with clear coefficients showing variable effects.

Classification

- **Decision Tree :**
  It effectively handles non-linear features.
- **XGBoost :**
  Superior performance & high efficiency.
- **Random Forest :**
  Strong robustness and higher accuracy.

Association

This is an unsupervised model, so it can uncover hidden associations and patterns between variables without requiring labelled data, helping us explore the underlying structure.

Limitations

Regression

- **XGBoost**
  Hard to interpret and more computationally expensive to train.
- **Linear Regression**
  Fails to capture nonlinear behavior, much lower predictive accuracy.

Classification

- **Decision Tree :**
  Lower stability and prone to overfitting.
- **XGBoost :**
  Sensitivity to outliers
- **Random Forest:**
  It sacrifices the intuitive rule-based explain ability.

Association

It only reveals correlations rather than causal relationships, so further experiments or modelling are needed to establish causality. In addition, it is quite sensitive to outliers, which means careful data cleaning and checking are required.